

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AP621
Subject Title	Advanced Electron Microscopy: Theory and Practice
Credit Value	3
Level	6
Pre-requisite/ Co-requisite/ Exclusion	Nil.
Objectives	The aim of this subject is to introduce concepts of electron microscopy and develop the ability of students to use the electron microscopes for material characterizations.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) Understand the fundamental theory of electron optics b) Understand the working principles of transmission electron microscopy(TEM) and scanning electron microscopy(SEM) c) Identify the functions of analytical TEM d) Master the practical skills of TEM and SEM operations e) Select appropriate methods for particular purposes of real material characterizations
Subject Synopsis/ Indicative Syllabus	Electron Optics: Fresnel diffraction; Fraunhofer diffraction; Fourier transform; Phase Contrast; Resolution; Aberration; Kinematic diffraction; Dynamical diffraction; Electron wave in crystals SEM: Secondary electron; Contrast; Resolution; Backscattered electron; EBSD; LEEM TEM: TEM compositions; Diffraction; Imaging; Spectroscopy Techniques for TEM and case study: HRTEM; SAED; CBED; Diffraction contrast; STEM; Lorentz Microscopy; Holography; EDS; EELS; in situ TEMs
Teaching/Learning Methodology	Lecture: Basic theory and knowledge behind all experiments will be systematically introduced in lectures. Class work and assignments related to the content of lectures will be used to enhance students learning. Laboratory session: Experiments are essential for students to relate the concepts to practical applications and they are exposed to hand-on experience and proper use of equipment and also analytical skills on interpreting experimental results.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d	e	
	1. Continuous assessment	60		√	√	√	√	
	2. Examination	40	√	√	√	√	√	
	Total	100 %						
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Assignments will strengthen the students' basic knowledge and the analytical skills to solve the problems related to this subject. Practical Tests is useful to assess students' experimental skills and knowledge learned from the lectures and lab works. Examination will review their understanding of the course assess their ability to solve problems. Hence, the proposed assessment methods are necessary to assess the intended learning outcomes.								
Student Study Effort Expected	Class contact:							
	▪ Lecture/Seminar		27 Hrs.					
	Other student study effort:							
	▪ Laboratory		12 Hrs.					
	▪ Self-study		81 Hrs.					
	Total student study effort		120 Hrs.					
Reading List and References	(1) D.B. Williams, C.B. Carter, Transmission Electron Microscopy, (Plenum Publishing Corporation, 1996). (2) J.C.H. Spence, High-Resolution Electron Microscopy (Oxford, 2013) (3) J. Cowley, Diffraction Physics (North Holland 1995) (4) J. Goldstein, et al, Scanning Electron Microscopy and X-ray Microanalysis (Springer 2003)							